System Project Requirement No. 1

ERD Documentation

AniMAL - A Web-Based Market Access Platform for Farmers

Direct Farm-to-Buyer Transaction Platform

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Database Management Systems (DBMS) - Systems Project Requirement

1. System Description

AniMAL is a web-based market access platform that connects farmers directly to buyers. Farmers publish produce listings with prices, stock levels, and quantities; buyers place orders against those listings and track order status. Administrators supervise the marketplace through order holds, account suspensions, retention holds, and audit logging. The system's Entity Relationship Diagram (ERD) models these entities and the 17 relationships that govern how farmers, buyers, and administrators interact.

2. Group Members

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3. Entity Descriptions

users -System accounts; the single sign-on identity for all roles (farmer, buyer, admin).

sessions -Active login sessions with device and IP fingerprint (VARCHAR(128) session id).

farmer_profiles -Verified seller records: farm name, location, valid ID, and verification workflow (keyed by user_id).

buyer_profiles -Shopper records: full name, contact, delivery and pickup preferences (keyed by user_id).

produce_categories -Product taxonomy for classifying listings.

quantity_units -Supported units of measure and conversion factors.

produce_listings -Sellable produce posts with price, stock, and lifecycle status.

listing_images -Photos attached to a listing; primary flag for thumbnails.

stock_adjustments -Audit trail of every stock change to a listing.

orders -Order header holding buyer, farmer, listing, delivery, amounts, and current status.

order_status_history -Immutable log of order state transitions with actor and reason.

admin_order_holds -Admin intervention flags that pause an order until released.

user_suspensions -Account suspension records and lift audit.

retention_holds -Legal/administrative holds preventing deletion of entities.

admin_audit_log -Before/after JSON snapshots of admin actions.

4. Relationship Summary (Cardinality)

Relationship	Cardinality	Notes
users : sessions	One-to-Many	A user may have many active sessions.

Relationship	Cardinality	Notes
users : farmer_profiles	One-to-Zero-or-One	A user (farmer role) may have one farmer profile.
users : buyer_profiles	One-to-Zero-or-One	A user (buyer role) may have one buyer profile.
farmer_profiles : produce_listings	One-to-Many	A farmer publishes many listings.
produce_categories : produce_listings	One-to-Many	A category groups many listings.
quantity_units : produce_listings	One-to-Many	A unit type is reused by many listings.
produce_listings : listing_images	One-to-Many	A listing has many photos.
produce_listings : stock_adjustments	One-to-Many	A listing has many stock change records.
buyer_profiles : orders	One-to-Many	A buyer places many orders.
farmer_profiles : orders	One-to-Many	A farmer fulfils many orders.
produce_listings : orders	One-to-Many	A listing is ordered many times.
orders : order_status_history	One-to-Many	An order transitions through many statuses.
orders : admin_order_holds	One-to-Many	An order may be held many times.
users : admin_order_holds (admin)	One-to-Many	Admin user places or releases many order holds.
users : user_suspensions (admin)	One-to-Many	Admin user suspends or lifts many accounts.
users : retention_holds (admin)	One-to-Many	Admin user applies placeholders that block deletion.
users : admin_audit_log (admin)	One-to-Many	Admin user generates many audit log entries.

5. Notation

The diagram uses Crow's Foot notation. A crow's foot (three prongs) marks the MANY side of a relationship; a single bar marks the ONE side. A circle on a side marks an optional (zero-or-one) participation. Audit-attribution foreign keys that are not drawn as lines (verified_by, released_by, lifted_by) appear as FK badges on the field; administrator actions (admin_order_holds, user_suspensions, retention_holds, admin_audit_log) are drawn as relationships from users.